



[SEP 2023] TEST 1 FSPK0022 JULY 2023-2024 16-9-2023

CALCULUS - foundation (Universiti Teknologi Malaysia)



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UNIVERSITI TEKNOLOGI MALAYSIA UTM

FOUNDATIONS

SEMESTER JULY 2023/2024

COURSE CODE : FSPK0022
COURSE NAME : FUNDAMENTALS OF COMPUTING
TIME : 5.30PM – 6.45PM
DATE : 19 SEPTEMBER 2023
VENUE : ARAS 1 PRZS UTMJB / PSZ UTMKL

INSTRUCTIONS TO THE STUDENTS:

This exam book consists of three (3) parts:

Part A :	10 Multiple Choice Question	10 Marks
Part B:	6 Structured Question	50 Marks
Part C:	1 Programming Question	10 Marks

TOTAL MARKS: 70 Marks

CARRYMARKS: 20%

ANSWER ALL QUESTIONS IN THE SPACES PROVIDED

Name	
Matric Number	
Name of Lecturer	
Section	

This examination book consists of 11 printed pages, including this page.

PART A: MULTIPLE CHOICE QUESTION (10 MARKS)

Answer all the questions and circle your answer in the space provided.

1. To translate a high-level language into machine codes, a computer needs a/an _____ program.
 - A. Assembler
 - B. Loader
 - C. Compiler
 - D. Linker

2. C++ is one of the _____ programming languages.
 - A. 1st generation
 - B. 2nd generation
 - C. 3rd generation
 - D. 4th generation

3. _____ is used to fix the _____, and _____ is used to fix _____.
 - A. Debugger, logical errors, compiler, syntax errors,
 - B. Debugger, syntax errors, compiler, logical errors
 - C. Linker, syntax errors, compiler, logical errors.
 - D. Debugger, syntax errors, Loader, logical errors.

4. The appropriate data type for a value of **12.50** is _____.
 - A. short
 - B. float
 - C. char
 - D. int

5. Which from the below statement is a correct pre-processor directive?

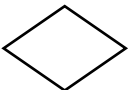
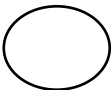
- A. `#include <iostream>`
Using namespace std;
- B. `#include <iostrem>`
using namespace std
- C. `#include<iostream>`
using namespace std
- D. `#include <iostream>`
using namespace std;

6. What is the answer for the following pseudocode?

```
1. Start
2. A = 12
3. B = 23
4. C = 10
5. C = 12 * 23
6. Display C
7. End
```

- A. 12
- B. 10
- C. 276
- D. 256

7. Which flowchart symbol below is suitable for the selection statement?

- A. 
- B. 
- C. 
- D. 

8. 17.86×10^3 would be written in E notation as
- A. 1786E3
 - B. 1.786E2
 - C. 17.86E3
 - D. 178.6e-2
9. Which of the following is the INVALID identifier?
- A. VARIABLE
 - B. variabl3
 - C. vari.able
 - D. variable_
10. Constant that begins with '0x' represents the number system for:
- A. Hexadecimal
 - B. Decimal
 - C. Binary
 - D. Octal

PART B: STRUCTURED QUESTIONS [50 MARKS]

Answer all the questions and write your answer in the space provided.

Question 1

Describe the term of Low-level language and the High-level language. Give TWO (2) examples for each of them.

(5Marks)

	Low-level Language	High-level language
Explanation		
Example		

Question 2

I. Write pseudocode based on the following table.

PNGK	Class
3.50 - 4.00	First Class Diploma
2.00 - 3.50	Second Class Diploma

(5Marks)

II. Convert the following pseudocode into a flowchart.

```
start
  get number
  pow = number * number
  if pow > 100 then
    print "A"
  else
    if number < 10 then
      print "B"
    else
      print "C"
    endif
  else
    print "D"
  endif
print "yes"
stop
```

(5Marks)

Question 3

- I. Identify and circle the error in the following statement. State the reason why such error occurs.

(7Marks)

	C++ Statement	Answer
a)	char letter, int num;	
b)	char 's';	
c)	char gender (m/f);	
d)	int marks = 86.5;	
e)	float price = RM345.50;	
f)	int kilometer = 1,250;	
g)	char grade, 'A', 'B', 'C'	

- II. State whether the following identifier (variables) are VALID or INVALID. If INVALID, state your reason.

Identifier	Valid/Invalid	Reason If Invalid
a) Weight2boys		
b) sumOfTWOnum		
c) _sum4num		
d) km_per_hour		
e) rate/hour		
f) xyz		

(6Marks)

Question 4

Write the equivalent C++ statement for the following algebraic equations. Use mathematic function where appropriate.

(6Marks)

a) $a^2 + b^2 = c^2$	b) $A = \pi r^2$
c) $S = \frac{a + b \times c}{b \times c + d}$	d) $d = (y2 - y1)^2 + (x2 - x1)^2$
e) $U = abcd + \frac{2}{6}$	f) $m = \frac{4(abc)}{2s}$

Question 5

Given the following program below:

1	#include <iostream>
2	using namespace std;
3	
4	// Task : Calculate the area of a circle
5	// Date : 19 September 2023
6	
7	int main()
8	{
9	float radius, area;
10	const float PI = 1.142;
11	
12	cout << "Please enter a radius";
13	cin >>radius;
14	
15	area = PI * radius * radius;
16	
17	cout << "The area of this circle is : " << area;
18	
19	return 0;
20	}

(6Marks)

Identify the following components by writing the line number of the statement.

Example: Input statement : Line 13

- | | | |
|-----------------------------|---|-------|
| a) The comment | : | _____ |
| b) Variable declaration | : | _____ |
| c) Preprocessor directives | : | _____ |
| d) Initialization statement | : | _____ |
| e) Output Statement | : | _____ |
| f) Expression | : | _____ |

Question 6

I. What is the value of the following expression?

	Expression	Value
a)	(10 >= 8 && 16 < 25)	
b)	(5 < 4 7 >=)	
c)	(9 == 9 && 0 < 4 2 > 3)	
d)	!(((4 + 3 > 5) && 7 < 8))	
e)	! (2*4 >= 3*2 10 != 5)	

(5Marks)

II. Given the value of the variable as follows:

int a = 4, b = 2, c = 3, d = 5;

What is the value of the following expression?

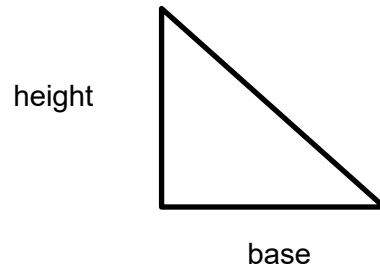
	Expression	Value
a)	(a + c < b + c)	
b)	(b < d c > a)	
c)	(a == b && c >= 5)	
d)	!(((c < d) && a < 9))	
e)	! (2*b > 10 12 >= d)	

(5Marks)

PART C: PROGRAMMING QUESTIONS [10 MARKS]

Answer all the questions and write your answer in the space provided.

Here is a diagram of a Triangle.



Write a C++ program to calculate the area of Triangle. The value of base and height are 5 and 4. All units of measurement are in cm and display the area of Triangles.

The area is given by the formula:

$$\text{area} = \frac{1}{2} \times \text{base} \times \text{height}$$

END OF QUESTION PAPER